## Exercise 3.15

1. A garment shop announces a flat 50% discount on every purchase of items for their customers. Draw the graph for the relation between the Marked Price and the Discount. Hence find
   (i) the marked price when a customer gets a discount of ₹3250 (from graph)
   (ii) the discount when the marked price is ₹2500
2. Draw the graph of $xy = 24$, $x, y > 0$. Using the graph find,
   (i) $y$ when $x = 3$ and (ii) $x$ when $y = 6$.
3. Graph the following linear function $y = \frac{1}{2}x$. Identify the constant of variation and verify it with the graph. Also (i) find $y$ when $x = 9$ (ii) find $x$ when $y = 7.5$.
4. The following table shows the data about the number of pipes and the time taken to till the same tank.

| **No. of pipes ($x$)** | 2 | 3 | 6 | 9 |
|---|---|---|---|---|
| **Time Taken (in min) ($y$)** | 45 | 30 | 15 | 10 |

   Draw the graph for the above data and hence
   (i) find the time taken to fill the tank when five pipes are used
   (ii) Find the number of pipes when the time is 9 minutes.
5. A school announces that for a certain competitions, the cash price will be distributed for all the participants equally as show below

| **No. of participants ($x$)** | 2 | 4 | 6 | 8 | 10 |
|---|---|---|---|---|---|
| **Amount for each participant in ₹ ($y$)** | 180 | 90 | 60 | 45 | 36 |

   (i) Find the constant of variation.
   (ii) Graph the above data and hence, find how much will each participant get if the number of participants are 12.
6. A two wheeler parking zone near bus stand charges as below.

| **Time (in hours) ($x$)** | 4 | 8 | 12 | 24 |
|---|---|---|---|---|
| **Amount ₹ ($y$)** | 60 | 120 | 180 | 360 |

   Check if the amount charged are in direct variation or in inverse variation to the parking time. Graph the data. Also (i) find the amount to be paid when parking time is 6 *hr;* (ii) find the parking duration when the amount paid is ₹150.



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